Parameter Analysis of 16 Psyche: Rotation Peroid and Spectral Class

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Abstract

The Bus-DeMeo taxonomic classification system aims to use a simple pipeline of 4 common photometric survey filters to quickly and easily classify asteroids in order to determine material composition and relation. Our calculated GRI slope of 5.18 ± 0.17 *%/100nm value sits directly within the known taxonomic class (X). We observe 16 Psyche, a well-known asteroid of high scientific value, over a span of approximately 48 hours, and infer its shape and rotational period. We also observe its shape over this span, and determine its rotational periods. We find that a primary rotation period of 4.2 hours is in excellent agreement with the empirical literature value of 4.196 h, validating the sensitivity of our periodogram and iterative fitting routines to the dominant signal. The physical validity of the full model remains constrained by data quality. We recommend a sampling rate of at least 30 data points per 4-hour rotation to satisfy the accuracy for the harmonics and interactions in the future.*

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1 History and Context

What is 16 Psyche and how did it form? Why is it significant to study this asteroid?
Asteroid 16 Psyche was discovered in 1852 by Italian astronomer Annibale de Gasparis. Being the 16th asteroid discovered, it orbits the sun in the outer part of the main belt and has an irregular potato-like shape. The asteroid is comparatively dense and measures approximately 278km, 238km, 168km on the axes of an asymmetric ellipse. Being the largest discovered M-type asteroid, it is packed with metals and sufficiently heavy to perturb the orbits of other asteroids. Recently, the asteroid was observed in radar, which discovered irregular patches of albedo nearly twice its average, suggesting highly non-uniform distribution of metal around and on the surface. The apparent correlation between the optical and radar albedo led to the hypothesis there there is a link in the process that creates regions of high metal content and surface terrain of high albedo.

The most well-supported hypothesis as a result of recent radar observations claim that Psych may be a differentiated object that causes radius-dependent (or other factors) difference in chemical composition and distribution. During cooling, it possibly experienced iron volcanism which resulted in its current distribution of metals.

In late 2023, NASA launched a probe at Psyche which will arrive in orbit around 2029 and will use its imagers, spectrometers, and other instruments to analyze the content of the asteroid. The mission is an undertaking to understand the composition and to possibly infer and model the processes which produce similarly exceptional asteroids, which would help further understand the early stage processes of proto-planets and asteroid formation.

Simple photometric surveys allow a researcher to do mass taxonomic analysis with minimal effort, making it possible to map asteroid material and mass (using density estimation) in large scale distributions within our solar system. Our study attempts to replicate and confirm previous characteristics.

2 Introduction and Theory

2.1 Physical Characteristics

2.1.1 $a > b > c$ with Uniform Albedo

Due to existing estimates suggesting a very high mass, we hypothesize that the asteroid is in hydrostatic equilibrium and is thus likely a triaxial ellipsoid. Assuming rotation about the principal moment of inertia (the minor axis c) and minimizing internal energy, we model the object with distinct semi-axes $a > b > c$.

We assume a uniform surface albedo, such that the observed brightness variation results purely from the changing geometric cross-section of the rotating body (shape-dominated light curve).

The expected light curve resulting from a triaxial ellipsoid rotating about its c-axis is:

$$\text{Flux}(t) = F_0 \cdot \pi \sqrt{\underbrace{a^2c^2\sin^2\theta\sin^2(\omega t)}_{\text{View of } a\text{-axis}} + \underbrace{b^2c^2\sin^2\theta\cos^2(\omega t)}_{\text{View of } b\text{-axis}} + \underbrace{a^2b^2\cos^2\theta}_{\text{View of Pole}}} \tag{1}$$

Where:

- t is the time.
- F_0 is a constant scaling factor (accounting for distance, albedo, and integration).
- a, b, c are the semi-axis lengths ($a > b > c$).
- θ is the Aspect Angle (angle between the rotation axis and line of sight).
- ω is the angular rotation frequency ($2\pi/T$).

The second-order Taylor series approximation reveals the harmonic structure:

$$F(t) \approx F_0\sqrt{\mu}\left[\underbrace{\left(1 - \frac{\Delta^2}{16\mu^2}\right)}_{\text{Mean Flux Shift}} + \underbrace{\frac{\Delta}{2\mu}\cos(2\omega t)}_{\text{Fundamental}} - \underbrace{\frac{\Delta^2}{16\mu^2}\cos(4\omega t)}_{\text{2nd Order Harmonic}}\right] \tag{2}$$

Where:

- $\Delta = \frac{c^2 \sin^2\theta(b^2-a^2)}{2} < 0$ (controls Amplitude)
- $\mu = \frac{c^2 \sin^2\theta(a^2+b^2)}{2} + a^2b^2\cos^2\theta$ (controls Mean Area)

Over short observation timespans, the Solar Phase Angle (α) which subtends the sun and earth is effectively constant and is absorbed into the amplitude and mean flux scaling.

For the final regression model, we treat the amplitudes and phases as free parameters to account for the unknown orientation:

$$F(t) = \bar{F} + A_1\cos\left(\frac{4\pi}{T}t + \phi_1\right) + A_2\cos\left(\frac{8\pi}{T}t + \phi_2\right) \tag{3}$$

Note: The fundamental frequency is 2ω (double-peaked), hence the angular terms $4\pi/T$ and $8\pi/T$.

2.1.2 $a = b > c$ with Uniform Albedo

If the asteroid is more symmetric on the major and intermediate axes, so that $a = b$, then the flux is no longer time-dependent:

$$F(t) = \bar{F} \tag{4}$$

Resulting in $\Delta = 0$ and a constant expected light curve.

2.1.3 $a = b > c$ with Inhomogeneous Albedo

Due to potential for an asymmetry of distribution of albedo on the surface, we postulate that the asteroid has spots of varying albedo, modeled as a brighter and darker side.

$$\text{Flux}(t) = \underbrace{F_0 \cdot A_{proj}}_{\text{Mean Flux}} \cdot \underbrace{[1 + \epsilon\sin(\theta)\cos(\omega t)]}_{\text{Albedo Modulation}} \tag{5}$$

Where ϵ is the albedo contrast factor, normalized by albedo mean. The flux is entirely sinusoidal. The fit will be:

$$\text{Flux}(t) = \bar{F} + A\cos(\frac{2\pi}{T}t) \tag{6}$$

2.1.4 $a > b > c$ with Inhomogeneous Albedo

In a more general case, the asteroid exhibits both geometric elongation and surface inhomogeneity. The observed flux is the product of the time-varying projected area and the effective surface albedo.

$$\text{Flux}(t) \propto A_{proj}(t) \times \rho_{eff}(t) \tag{7}$$

Substituting the Taylor approximations from the previous models:

1. The Shape function oscillates primarily at 2ω (due to the symmetry of the ellipsoid).
2. The Albedo function oscillates at ω (due to the rotation of a single surface feature).

Multiplying these terms induces heterodyning. The interaction is proportional as such:

$$\cos(2\omega t) \cdot \cos(\omega t) \to \cos(\omega t) + \cos(3\omega t) \tag{8}$$

The resulting light curve is complex and asymmetric, containing power at ω, 2ω, and 3ω.

- ω: Dominated by the Albedo spot (Light/Dark side).
- 2ω: Dominated by the Ellipsoidal Shape (Elongation).
- 3ω: Interaction

For the regression model, we expand the Fourier series to include these interaction terms:

$$F(t) = \mu + A_1\cos\left(\frac{2\pi}{T}t + \phi_1\right) + A_2\cos\left(\frac{4\pi}{T}t + \phi_2\right) + A_3\cos\left(\frac{6\pi}{T}t + \phi_1 + \phi_2\right) \tag{9}$$

Note: If A_1 (Spot) and A_3 (Mixing) are negligible, the object collapses back to the pure Shape Model (A_2). If A_2 (Shape) is negligible, it collapses to the pure Albedo Model (A_1).

2.1.5 $a = b > c$ with Precession

We consider the case where the asteroid is a uniform oblate spheroid undergoing stable free precession (wobble) about its angular momentum vector.

As mentioned above, the pure rotation of the oblate spheroid causes constant flux, but precession causes the Aspect Angle θ to oscillate which in turn varies the cross-sectional.

The instantaneous aspect angle varies according to the precession frequency Ω_p:

$$\cos\theta(t) \approx \cos\beta + (\sin\beta\sin\gamma)\cos(\Omega_p t) \tag{10}$$

Where γ is the small wobble angle and β is the mean inclination of the angular momentum vector.

The fit will be:

$$F(t) \approx \bar{F} + A\cos(\Omega_p t) \tag{11}$$

2.1.6 $a = b > c$ with Precession and Inhomogeneous Albedo

This system is characterized by two distinct frequencies:

1. ω_s: The Spin Frequency (rotation about the body axis).
2. Ω_p: The Precession Frequency (wobble of the body axis).

The precession motion modulates the amplitude of the spot's signal, creating an AM effect.

$$F(t) \approx \bar{F}\left[1 + A_{prec}\cos(\Omega_p t) + A_{spin}\cos(\omega_s t)\cdot(1 + m\cos(\Omega_p t))\right] \tag{12}$$

For the regression:

$$\begin{aligned} F(t) \approx \bar{F} &+ \underbrace{A_{prec}\cos(\Omega_p t + \phi_p)}_{\text{Shape Change}} \\ &+ \underbrace{A_{spin}\cos(\omega_s t + \phi_s)}_{\text{Mean Spot Rotation}} \\ &+ \underbrace{A_{mix}\left[\cos((\omega_s - \Omega_p)t + \phi_s - \phi_p) + \cos((\omega_s + \Omega_p)t + \phi_s + \phi_p)\right]}_{\text{Sidebands}} \end{aligned} \tag{13}$$

Due to our undersampled data, the most general model which includes precession and varying albedo on a triaxial ellipsoid is excluded due to worries of overfitting.

2.2 Spectral Characteristics

The Bus-DeMeo taxonomic classification system aims to use a simple pipeline of 4 common photometric survey filters to quickly and easily classify asteroids in order to determine material composition and relation. Using color indices based on reflectance values, calculated from the absolute magnitudes of the objects of interest under the 4 different filters (g', r', i', z'), one can easily determine the material composition of the asteroid of interest.

These classes represent different material composition. For example, C-type refers to "Carbonaceous" asteroid, mostly consisting minerals high in carbon content. S-type refers to "Siliceous" stony asteroids, mostly consisting of silicates. X-type refers to "Metallic" asteroids, mostly consisting of metals. This is the least common of the 3 major classes, with less than 8% of asteroids being classified as X-type. There are further classifications, but they are not relevant. The asteroid of interest for this paper, 16 Psyche, is an X-type, consisting largely of iron and nickel.

Below is a table detailing the spectral characteristics under the classification system:

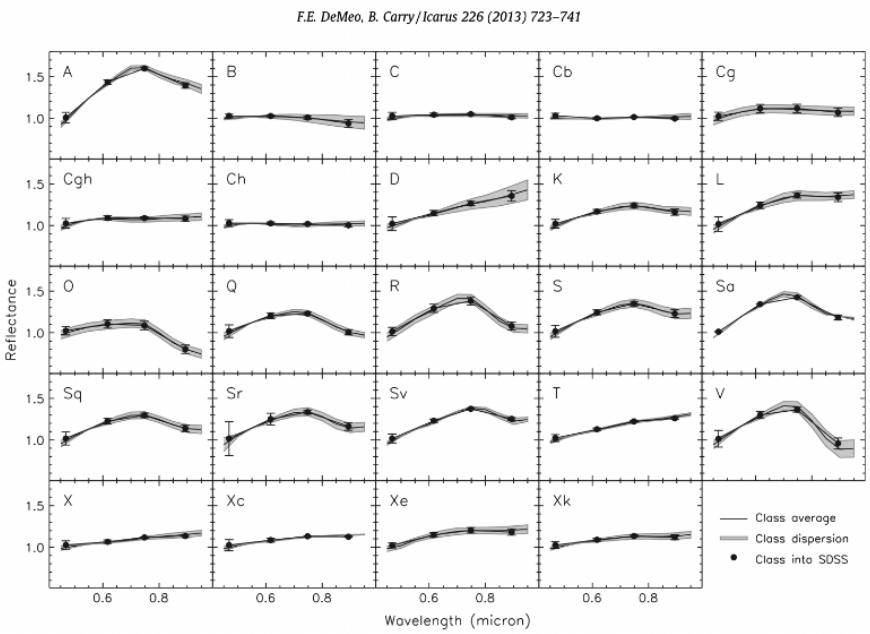


Figure 1: DeMeo, Carry. "Average Bus-DeMeo (DeMeo et al. 2009) spectra converted to SDSS colors used to define the classification boundaries. The black dots (with 1 standard deviation from the mean plotted) represent the average Bus-DeMeo spectra converted into SDSS colors. The u' filter is extrapolated from the data because the spectra do not cover those wavelengths (the u' filter is not used in the classification of SDSS data, however). The gray background plots the average spectrum plus one sigma for comparison with the colors. Because the Cg, O, and R classes are defined by a single object the standard deviation is set to 0.1."

3 Data Collection and Procedure

In the most idea scenario, the time-series observations would be conducted on the exact same sky conditions. That is, similar:

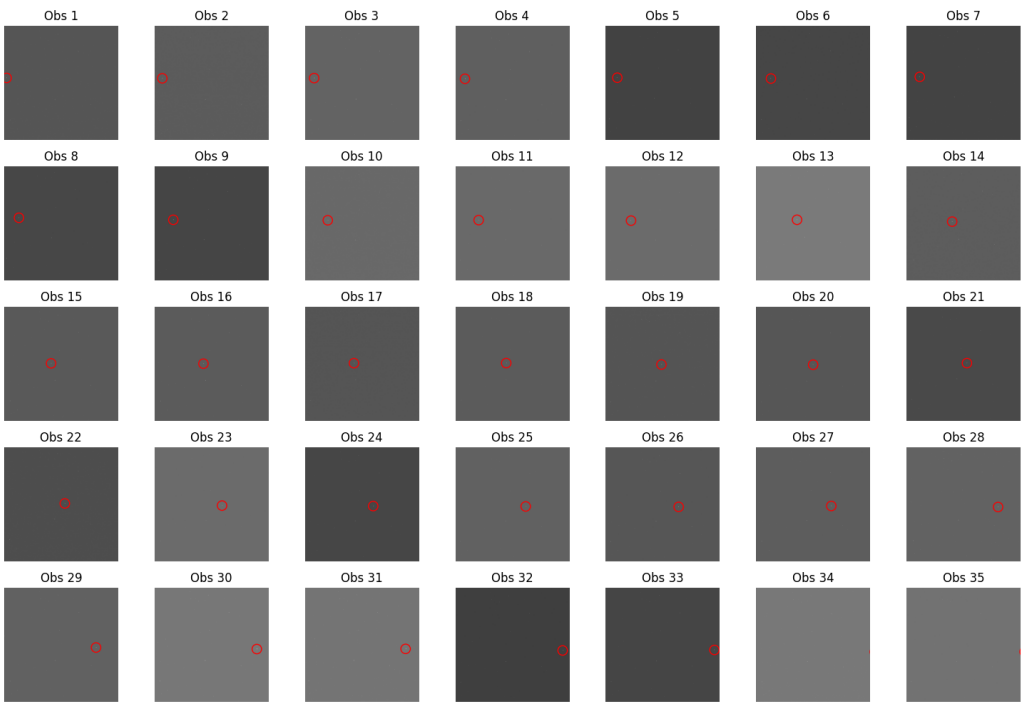
- Moon Phase
- Angle from Moon to Target
- Time of Day
- Location
- Airmass
- Weather and Seeing

However this is not the case: there is a queue for observation, and the system automatically chooses the best location based on its own optimization parameters. The best method then is to normalize all fluxes to the same scale from all images is to identify a set of stars used for reference across all observation frames. The frames will be normalized based on this. As such, a significant portion of our procedure is to obtain an accurate measurement of the desired normalization coefficient.

Sidereal coordinates were generated by NASA's Horizons system, and the apparent angular motion was used to calculate the timespan such that the 30' x 30' camera frame used by the QHY600 from LCO's network was aimed at the middle of the asteroid's expected transits. The estimated rotation peroid from previous measurements is 4.2 hours. We aimed our exposure times to achive a > 100 SNR on all filters. 3 (filters SDSS g,r,i) x 50 (frames) were requested with cadence 1 hour and jitter 1 hour in hopes that the under-sampling would be alleviated by the randomness of the time allotment from the jitter. However, only 50 frames requests on the g-band went through and 36 of the 50 were successful. Despite the setback, we proceeded.

Spectral Analysis: Since multiple bands are required for spectral analysis, separate observations were made. Requests with the same integration times were completed on the SDSS g,r,i in a 9-minute time-span. This part of the research was handled by my partner, who approached the calculations from a different systematic perspective.

The following image presents the transit across the 36 observations. The RA,DEC coordinates were mapped to the corresponding images' X,Y grid using the WCS information embedded in the .fits header files.



The following displays the data extraction and processing procedure:

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3.1 BANZAI Calibration Pipeline

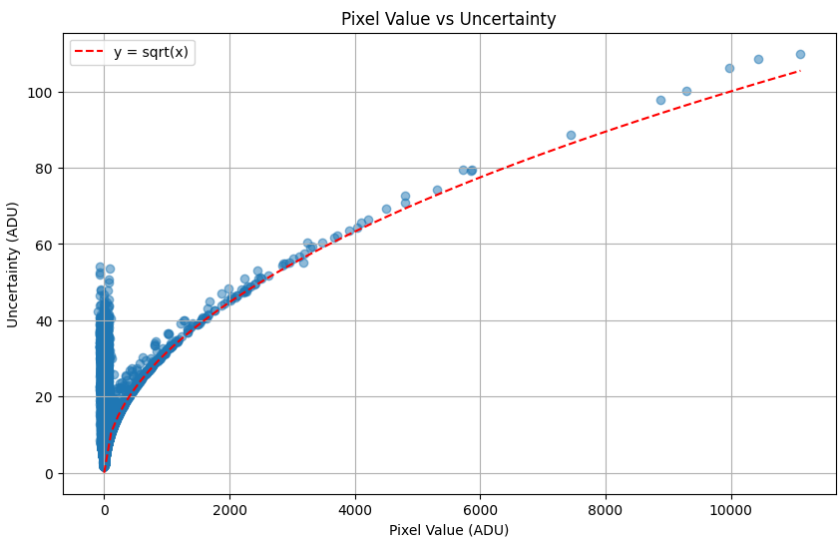
Beyond the simple exposure, the LCO telescopes take standard calibration frames. The following are applied to the image before released:

- Bad-pixel masking
- Bias subtraction
- Dark subtraction
- Flat field correction
- Source extraction (SEP) (obtains selected catalog data)
- Astrometric calibration (for WCS information)

Finally, the .fits files are release with the extensions: Scientific **SCI**, Bad Pixel Map **BPM**, Error **ERR**, Catalog **CAT**.

Crucially, the **ERR** extension contains the propagated cumulative error on all frames, which includes the read noise as stated on the website, mentioned elsewhere is that it also includes the Poisson and dark current noise, containing "pixel-by pixel uncertainties". Note that the gain is unity.

To confirm we have the correct information, we plotted the first frame's counts vs their corresponding errors. The Poisson noise, dominant at high N, is proportional to square root.



We expect the pixel array to contain the following information:

$$\sigma_{xy} = \sqrt{N_{xy} + N_{dark} + N_{read}^2 + (\cdots)_{xy}} \tag{14}$$

- N is the total source flux (e)
- $\tau = 2$ is the integration time (seconds)
- $N_{read} \approx 3.2$ is the read noise (e/pixel) (tends to vary across frames)
- $i_{DC} \approx 0.01$ is the dark current (e/pixel/s) @ 200K (constant)
- $N_{dark} = i_{DC} \cdot \tau$
- The rest, dependent on the x,y pixel comes from propagation of the other calibration frames

The **CAT** extension contains the background-subtracted flux automated by `SEP` using the Kron photometry method. The flux from 3 reference stars are chosen within half a magnitude of each other, along with the flux of the asteroid on each filter.

The Kron radius, the major axis (a), and the minor axis (b) are given in the data headers. For the BANZAI pipeline, a value of 2.5 is used for the Kron factor. Errors for spectral classification were manually calculated as such:

$$\sigma = \sqrt{N + n_{pix}(N_\beta + N_{dark} + N_{read}^2)} \tag{15}$$

$$n_{pix} = \pi \cdot \text{kron radius}^2 \cdot \text{kron factor}^2 \cdot a \cdot b \tag{16}$$

$$N_\beta = F_\beta \cdot A \cdot \epsilon \cdot Q_e \cdot \Omega \cdot \tau \tag{17}$$

The following were used for the values above:

- n_{pix} is the number of pixels in the photometric aperture.
- $F_\beta \approx 0.01$ is the background flux from the sky (photons/s/cm^2/arcsec).
- $A = \pi \cdot 0.35^2 \approx 0.38$ is the telescope area (cm^2).
- $\epsilon = 0.5$ is the telescope efficiency.
- $Q_e = 0.9$ is the quantum efficiency of the detector.
- $\Omega = 0.74$ is the pixel size (arcsec).
- $N_{read} = 1$ is the read noise (e/pixel).

Note that SNR is calculated as $SNR = N/\sigma$

3.2 Spectral Classification

Instrumental magnitudes were converted to the standard system using reference stars. The uncertainty in magnitude (σ_m) was derived from the flux uncertainty (inverse SNR) using error propagation:

$$\sigma_f = \sqrt{\left(\frac{\delta f}{\delta x}\sigma_x\right)^2 + \left(\frac{\delta f}{\delta y}\sigma_y\right)^2 + \dots} \tag{18}$$

With:

$$m = -2.5 \log_{10}\left(\frac{F_{ast}}{F_{ref}}\right) + m_{ref} \tag{19}$$

so:

$$\sigma_m = \sqrt{\left(-2.5\frac{1}{F_{ast}\ln(10)}\sigma_{ast}\right)^2 + \left(-2.5\frac{1}{F_{ref}\ln(10)}\sigma_{ref}\right)^2} \tag{20}$$

$$\sigma_m = 2.5\frac{1}{\ln(10)}\sqrt{\left(\frac{1}{\mathrm{SNR}_{ast}}\right)^2 + \left(\frac{1}{\mathrm{SNR}_{ref}}\right)^2} \tag{21}$$

Because the GAIA database was down, the SNR for the reference stars could not be obtained, however this is not a major issue because the uncertainty from our measurements will greatly outweigh that of the GAIA survey.

3.2.1 Reflectance Conversion

To determine the spectral slope, we converted apparent magnitudes into normalized reflectance (R_λ). We normalized all values to the g-band ($\lambda = 0.47\mu m$). The reflectance at filter λ is given by comparing the color of the asteroid to the color of the Sun:

$$R_f = 10^{-0.4\left[(m_f - m_g) - (C_{\odot,f})\right]} \tag{22}$$

Where $(m_f - m_g)$ is the measured asteroid color and $C_{\odot,f}$ is the solar color index for that filter pair. By subtracting out the spectral differences of the sun's light, we can get reflectance data dependent on purely the asteroid. This is important for comparing to reflectance data of minerals on Earth, allowing the classification system to be created in the first place. From *DeMeo*, $(m_r - m_g) = -0.45)$ and $(m_i - m_g) = -0.55)$ The propagated error for reflectance σ_{R_f} is:

$$\sigma_{R_f} = \ln(10) \cdot 0.4 \cdot R_f \cdot \sqrt{\sigma_{m_f}^2 + \sigma_{m_g}^2} \tag{23}$$

3.2.2 Weighted Linear Fit (Spectral Slope)

We determined the spectral slope (S) by performing a weighted linear least-squares fit to the normalized reflectance data points vs. wavelength (λ). We minimized the chi-squared statistic:

$$\chi^2 = \sum_i w_i (y_i - (S\lambda_i + c))^2 \tag{24}$$

3.2.3 Slope Uncertainty

Since asteroid spectra may contain intrinsic curvature (such as broad absorption features) that deviates from a linear model, the standard error of the regression derived from residuals often overestimates the observational uncertainty.

Instead, we calculate the formal uncertainty of the slope (σ_S) based purely on the propagation of photometric errors. In a weighted least-squares formulation, the variance of the slope is determined by the weights ($w_i = 1/\sigma_{R,i}^2$) and the distribution of wavelengths:

$$\sigma_S = \frac{1}{\sqrt{\Delta}} \tag{25}$$

where Δ is the determinant of the weight matrix, given by:

$$\Delta = \sum_i w_i \sum_i w_i \lambda_i^2 - \left(\sum_i w_i \lambda_i \right)^2 \tag{26}$$

This formulation ensures that the error in the spectral gradient reflects the signal-to-noise ratio of the photometry rather than the deviation of the asteroid's spectral shape from a perfect line.

Where the weights $w_i = 1/\sigma_{R,i}^2$. The resulting slope S represents the change in reflectance per micron. This was converted to the units of spectral gradient (%/100nm) for taxonomic classification.

3.3 Reference Stars Flux Extraction

3.3.1 Identification

To obtain the normalization factor, the USNO UCAC5 catalog data was chosen and its stars were identified on all frames, choosing a small range of brightnesses within 1 magnitude. We assume that the chosen stars have the same emission spectra around the g-band range and the atmospheric extinction in this range is constant.

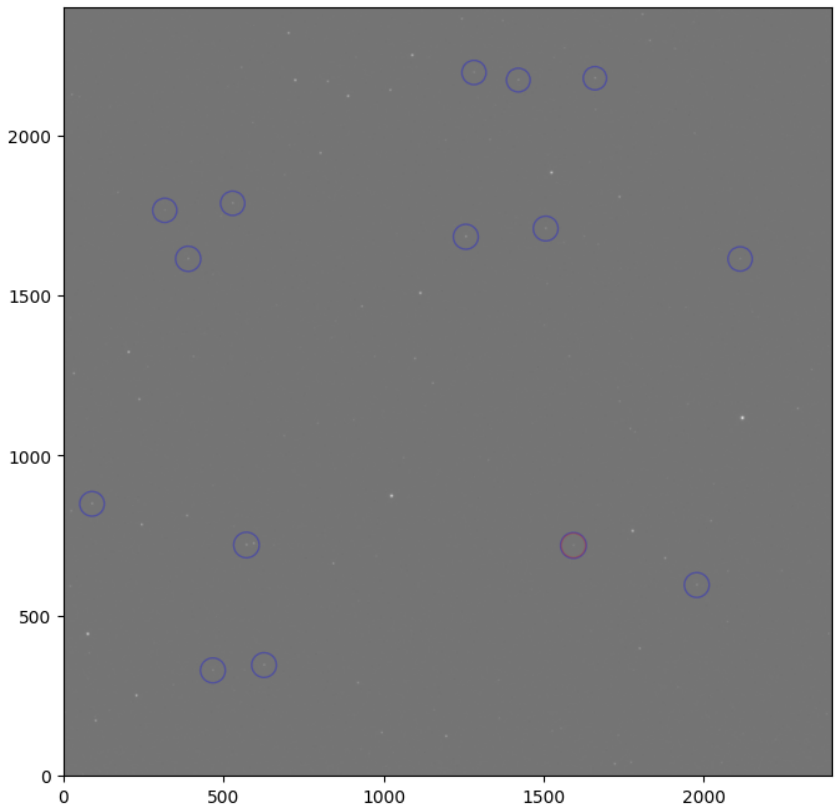


Figure 2: 30th Observation, the star in the purple circle has the highest σ in the gaussian fitting

3.3.2 Gaussian Fitting for Centroid and Radius

We apply a 2D elliptical gaussian fitting on all reference stars to extract the centroid position and a σ in image pixel spread (PSF). Across all reference stars, the highest σ is chosen as a base to calculate the aperture and annuli radii. The fixed area simplifies and reduces systematic error. The aperture is chosen to have $3\sigma_{worst}$, the inner annulus ring $5\sigma_{worst}$, and outer ring $7\sigma_{worst}$.

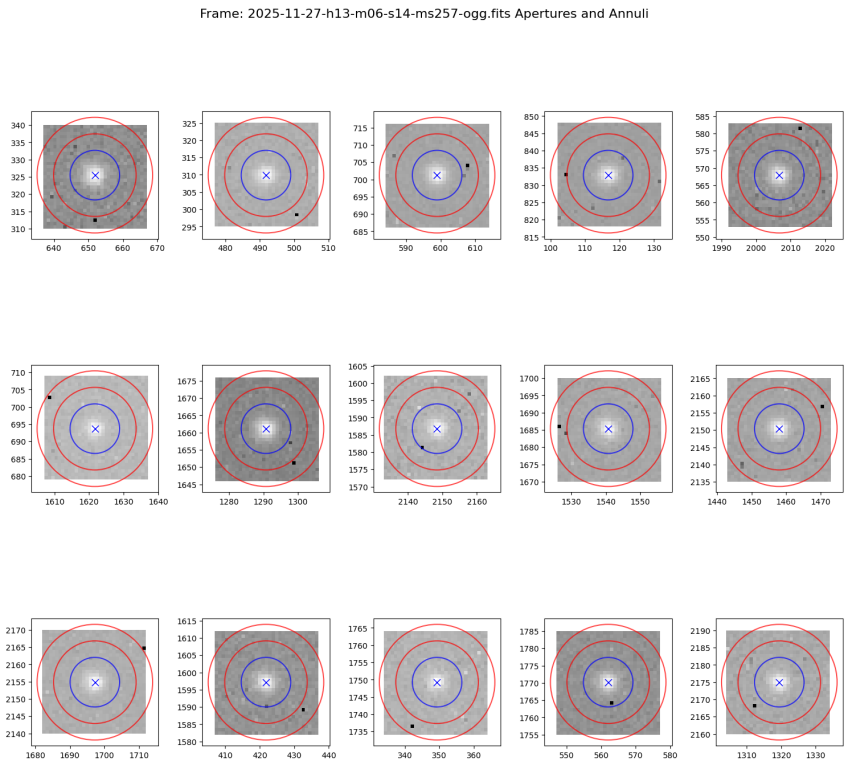


Figure 3: 30th Observation, visualized apertures and annuli

3.3.3 Photometry

Aperture photometry is conducted using the `astropy` and `photutils` python libraries. The flux is summed exactly on the shape of the apertures so that if only a fraction of the pixel is contained in the aperture, then that fraction of flux and the pixel uncertainty is taken, as such:

$$N_{ap} = \sum_{(x,y)\in A_{ap}} N_{x,y}, \qquad \sigma_{ap} = \sqrt{\sum_{(x,y)\in A_{ap}} \sigma_{x,y}^2} \tag{27}$$

To estimate the flux solely from the star, first we obtain the 3σ-clipped median as an estimate of the mean from the annulus ring. We assume the background pixels to be independently and identically random, where the Poisson noise is sufficiently high enough to approximate a gaussian. The sample median is used to calculate the Median Absolute Deviation (MAD), which is then used to estimate the standard deviation. For background pixels x_i in the annulus:

$$MAD = med(|x_i - \tilde{x}|), \qquad \tilde{x} = med(x_i) \tag{28}$$

$$\hat{\sigma}_{bkgpixel} = \frac{\text{MAD}}{\Phi^{-1}(0.75)} = \frac{\text{MAD}}{0.6745} \tag{29}$$

The standard deviation estimates the error of the background level.

$$\hat{\sigma}_{bkg} \approx \sqrt{\frac{\pi}{2}} \times \frac{\hat{\sigma}_{bkgpixel}}{\sqrt{A_{bkg}}} \tag{30}$$

The background value is multiplied by the area of the inner aperture to approximate the total background flux near the star.

$$N_{bkgest} = A_{ap} \times \tilde{x}, \qquad \hat{\sigma}_{estbg} = A_{ap} \times \hat{\sigma}_{bkg} \tag{31}$$

Then the estimate for the flux solely from the star is:

$$N_{star} = N_{ap} - N_{bkgest}, \qquad \sigma_{star} = \sqrt{\sigma_{ap}^2 + \sigma_{estbg}^2} \tag{32}$$

3.4 Norm Coefficient Calculation

To systematically remove the effect of the seeing conditions from the asteroid integrated flux, we need to normalize all calculations to a standard. We pool the ratios of the brightnesses of each reference star on all images to calculate the scaling factor that a frame should take to scale to standard. We assume that the atmospheric extinction effects are a mostly multiplication linear scalar for each frame.

To calculate the normalization coefficient of any one frame, we will weigh the measurements by their variance reciprocal as standard practice, as that is the signifier for their confidence. Specifically, the median is chosen instead of any observation in general to lower the propagated error.

For index i on frames, and j on reference stars on the reduced ADU count (N_{star}) observations x_{ij}:

$$R_i = \frac{\sum_j (w_{ij} \cdot \frac{x_{ij}}{\tilde{x}_j})}{\sum_j w_{ij}}, \qquad \tilde{x}_j = \text{median}(\{x_{ij}\}) \tag{33}$$

w_{ij} should be the variance reciprocal of its product partner (the ratio)

From earlier, the uncertainty of the sample median is on the left, and so the median of a reference star across multiple frames is on the right:

$$\sigma_{med} \approx \sqrt{\frac{\pi}{2N}}\sigma, \qquad \tilde{\sigma}_j \approx \sqrt{\frac{\pi}{2N_{frames}}}\bar{\sigma}_j \tag{34}$$

Where N_{frames} is the number of frames, and $\bar{\sigma}_j$ is the estimator of the standard deviation of the reference star by the average of its errors across the frames. As we divide*, we obtain the propagated error of the reference star ratios.

$$\frac{Var(\frac{x_{ij}}{\tilde{x}_j})}{\left(\frac{x_{ij}}{\tilde{x}_j}\right)^2} = \left(\frac{\sigma_{ij}}{x_{ij}}\right)^2 + \left(\frac{\tilde{\sigma}_j}{\tilde{x}_j}\right)^2 = \left(\frac{\sigma_{ij}}{x_{ij}}\right)^2 + \left(\frac{\bar{\sigma}_j}{\tilde{x}_j}\right)^2 \frac{\pi}{2N}, \qquad w_{ij} = \frac{1}{Var(\frac{x_{ij}}{\tilde{x}_j})} \tag{35}$$

As such, the variance of the weighted frame ratio simplifies to:

$$\sigma_{R,i}^2 = Var(R_i) \approx \frac{1}{\sum_j w_{ij}} \tag{36}$$

*Because N_{frames} is large, the covariance from the single observation and the median is minimal and ignored.

3.5 Asteroid Flux and Scaling

In the same manner, we apply the gaussian fitting on the asteroids to extract the pixel centroid and standard deviations. The worst σ is chosen for radii, and aperture photometry with background subtraction is performed. The radii are as follows: $\sigma_{worst} \cdot (5,7,9)$ for the aperture, inner annulus, and outer annulus radii respectively.

Obtaining $N_{ast,i}$,$\sigma_{ast,i}$ on frame i, we scale all asteroid data respectively:

$$F_{final,i} = \frac{F_{ast,i}}{R_i} \tag{37}$$

$$\frac{Var(F_{final,i})}{F_{final,i}^2} = \frac{Var(N_{ast,i})}{N_{ast,i}^2} + \frac{Var(R_i)}{R_i^2} \tag{38}$$

Where $\sigma_{final,i} = \sqrt{Var(F_{final,i})}$.

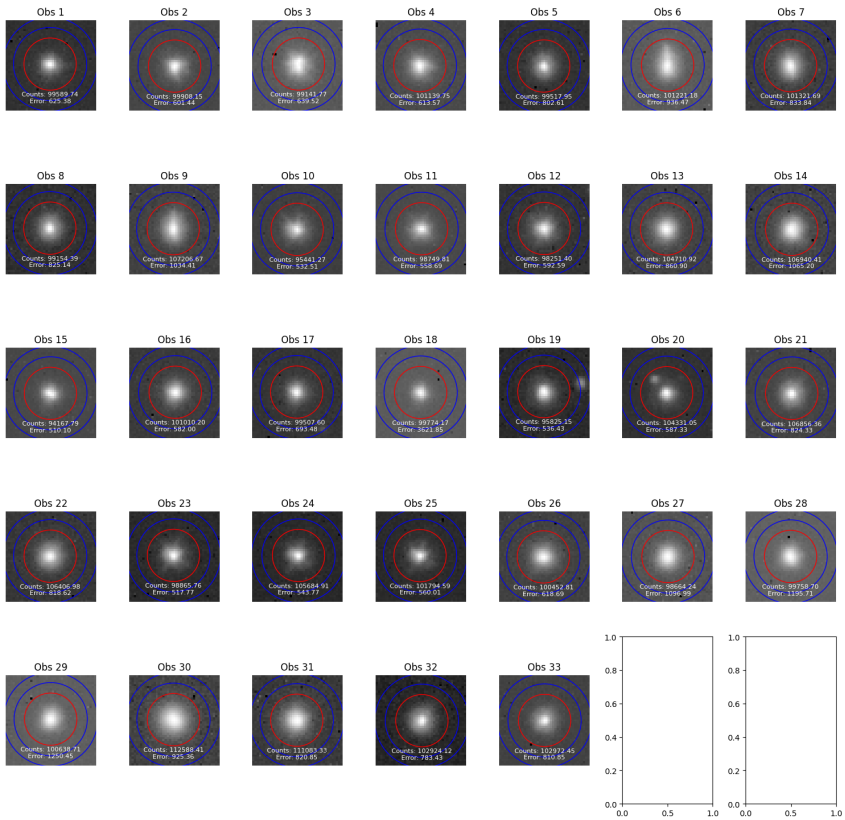


Figure 4: Scaled counts with error of asteroid on all reference frames
Last two do not capture asteroid. Best SNR$\approx$ 200, worst SNR$\approx$ 30

Filter	Exp (s)	Flux (Ast)	Flux (Ref 1)	Flux (Ref 2)	Flux (Ref 3)	Ref 1 Mag	Ref 2 Mag	Ref 3 Mag	Ast Mag	Ast SNR
g	2.0	92946.2 ± 305.3	122892.8 ± 351.0	61521.6 ± 351.0	79813.1 ± 283.0	9.724	10.538	10.233	10.061 ± 0.003	304.4
r	2.0	80617.9 ± 284.4	82903.8 ± 307.7	126311.1 ± 307.7	101530.0 ± 330.1	9.520	9.071	9.308	9.556 ± 0.003	283.5
i	2.0	37471.3 ± 194.1	33202.7 ± 196.9	88212.8 ± 196.9	56889.6 ± 247.9	9.470	8.418	8.889	9.343 ± 0.004	193.1

Figure 5: Flux and Magnitude data from the imagery and calculated as detailed above. Reference Star magnitudes in each SDSS filter is given by the GAIA survey. No error was assumed on this data as the error is much lower than our data and thus negligable

4 Analysis

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4.1 Spectral Characteristics

The table below presents the aggregated data: It was unnecessary to convert asteroid magnitude to solar system magnitude (which uses 1 AU) because the difference under the logarithm negates the reference.

$$m_i = -2.5 \log_{10} \left(\frac{F_{ast}}{F_{ref_i}} \right) + m_{ref_i} \tag{39}$$

And then:

$$m = \frac{1}{3} \sum_{i=1}^{3} m_i \tag{40}$$

By using the reference stars and their absolute magnitudes as calculated in the GAIA sky survey, we did not need to account for differential atmospheric absorption of the different bands. We did not need to worry about any other differences in sensitivity as well because by normalizing that would not be necessary. Differences in error are taking into account in the raw flux value of the SNR equation as well.

We achieved a very high signal to noise ratio for the observations, in excess of 100. This was possible due to the very low magnitude of asteroid 16 Psyche, with the highest magnitude of 10. For the reference stars, the GAIA database was accessed in order to get the absolute magnitude of each of the reference stars.

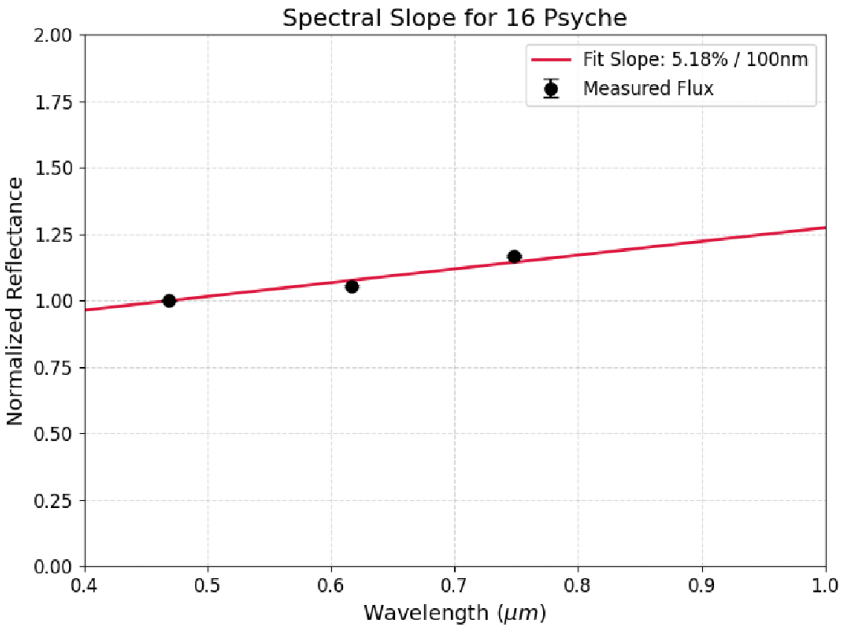

Figure 6: Normalized reflectance spectrum of the target asteroid. The calculated slope is $(5.18 \pm 0.17)\%/100nm$

Class	Slope (%/100 nm)		z′–i′	
	(min)	(max)	(min)	(max)
A	21.5	28.0	−0.265	−0.115
B	−5.0	0.0	−0.200	0.000
C	−5.0	6.0	−0.200	0.185
D	6.0	25.0	0.085	0.335
K	6.0	11.0	−0.075	−0.005
L	9.0	25.0	−0.005	0.085
Q	5.0	9.5	−0.265	−0.165
S	6.0	25.0	−0.265	−0.005
V	5.0	25.0	−0.665	−0.265
X	2.5	9.0	−0.005	0.185

Figure 7: Arrows point to classes within our fit.

The asteroid falls into multiple taxonomic complexes: C, V, and X. 16 Psyche is an X-class asteroid and our data is consistent with this. Without the z' data, we were unable to determine the z'-i' color index which would have made further class refinement possible.

4.2 Physical Characteristics

To begin, we take a look at the adjusted counts data and visualize the error bars.

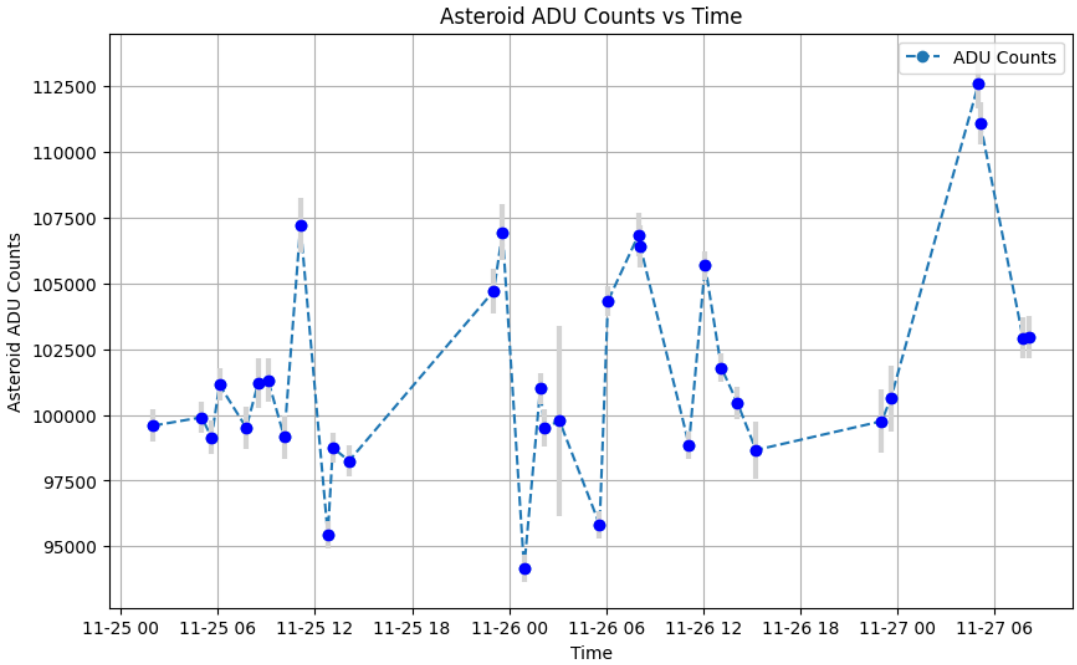

Figure 8: Adjusted counts data with $\pm 1\sigma$ error bars

Due to the under-sampling of our data, the confidence of the following fits are low. With nonlinear least-squares optimization in `scipy` with inverse variance weights for the errors, we obtain the following:

4.2.1 Simple Sinusoid

$$y \sim \mu + A \sin(2\pi/T \cdot t + \phi_1) \tag{41}$$

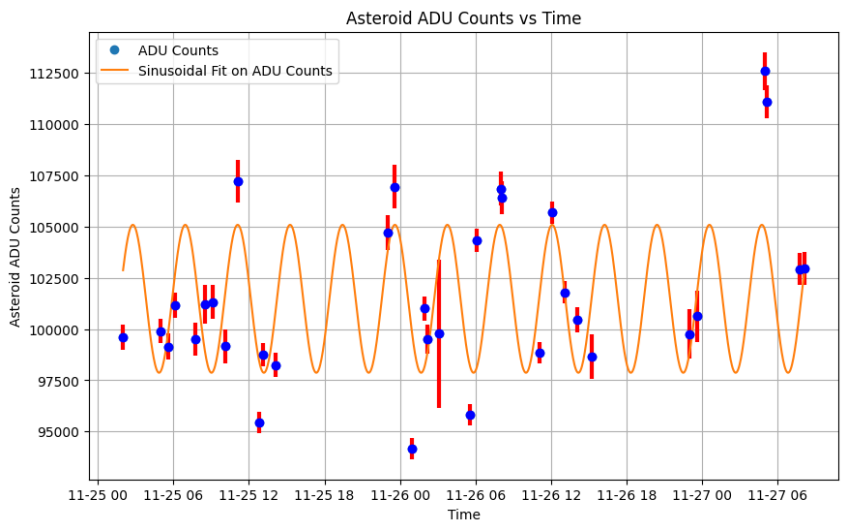

Figure 9: $T \approx 4.16h$

4.2.2 Double Sinusoid with Harmonic Constraint

The following approximates the Triaxial ellipsoid model with varying albedo. A linear term is included as the radial distance was closing during the observation period.

$$y \sim \mu + A\sin(2\pi/T \cdot t + \phi_1) + B\sin(4\pi/T \cdot t + \phi_2) + a \cdot t \tag{42}$$

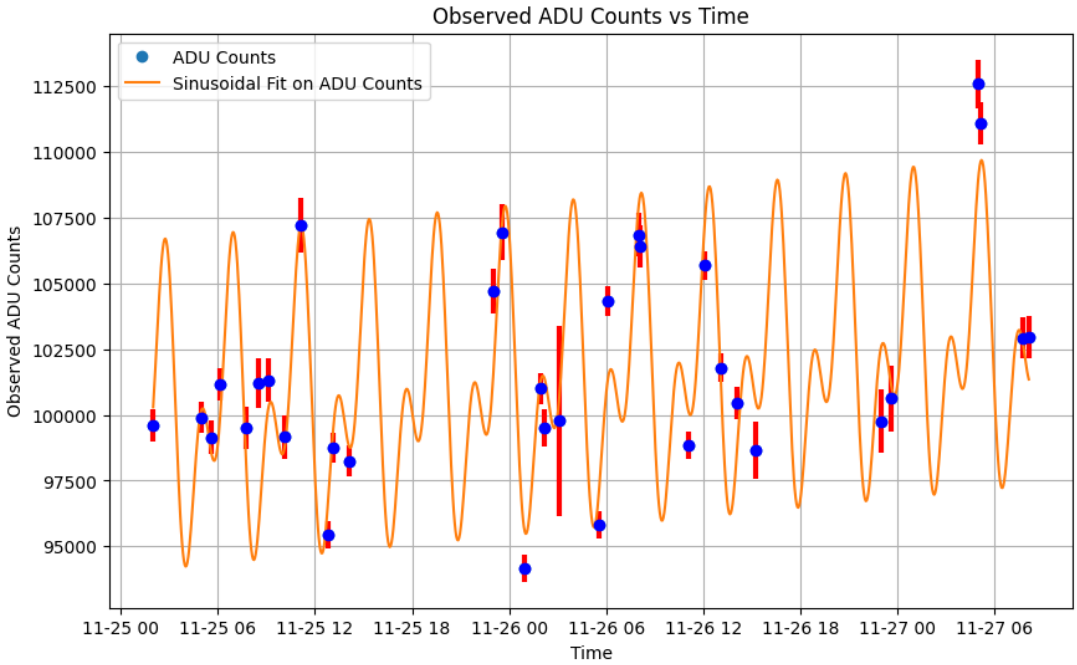

Figure 10: $\chi^2 \approx 183, T \approx 4.203h$

4.2.3 Best: Triaxial Ellipsoid with Varying Albedo

Realistically, the third term is an interaction between the spot and shape model, and its phase offset is related by sum we impose the constraint $\phi_3 = \phi_1 + \phi_2$

$$y \sim \mu + A\sin(2\pi/T \cdot t + \phi_1) + B\sin(4\pi/T \cdot t + \phi_2) + C\sin(6\pi/T \cdot t + \phi_3) + a \cdot t \tag{43}$$

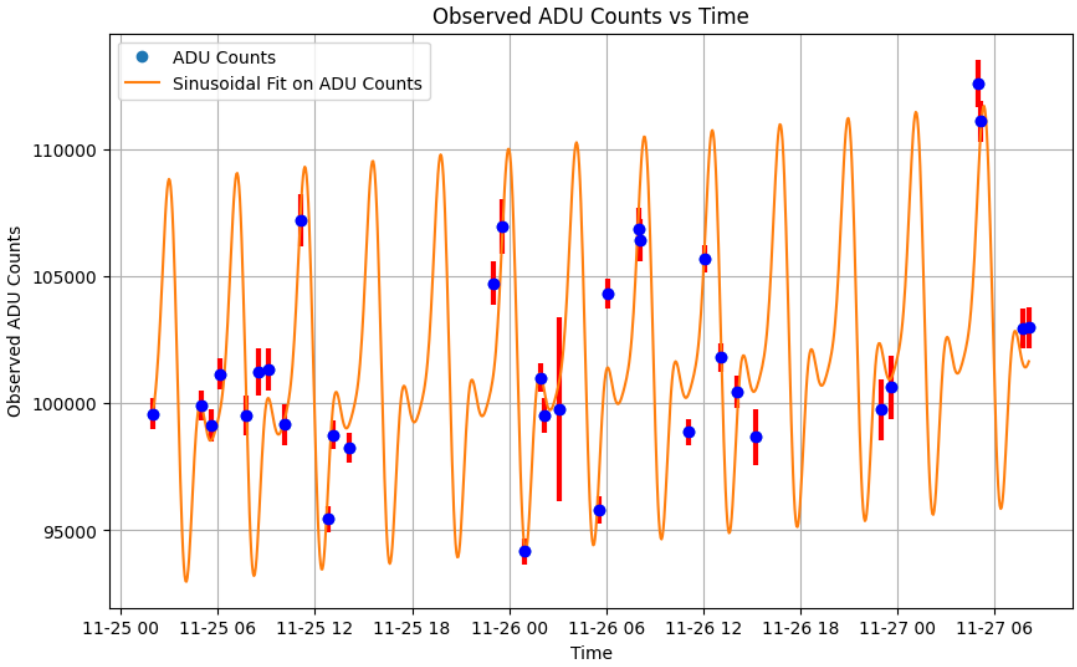

Figure 11: $\chi^2 \approx 169$,$T \approx 4.195h$,$A \approx 4385$,$B \approx 4015$,$C \approx -1386$

4.2.4 Oblate Spheroid with Precession and Varying Albedo

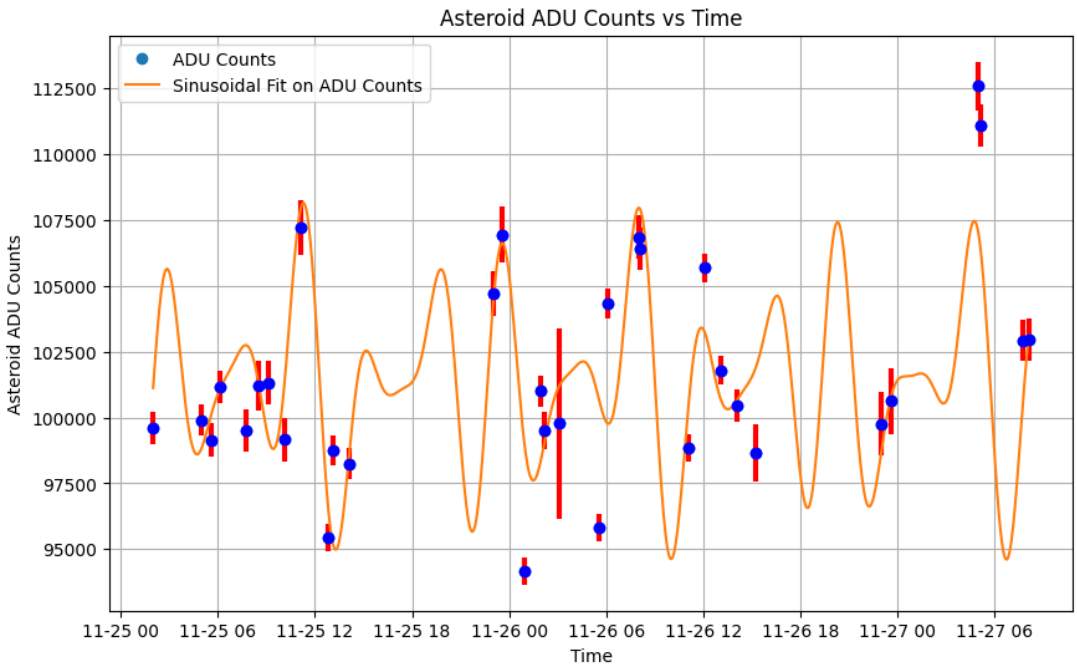

Figure 12: $\chi^2 \approx 471, T_{spot} \approx 4.14h$,$T_{prec} \approx 10.62h$

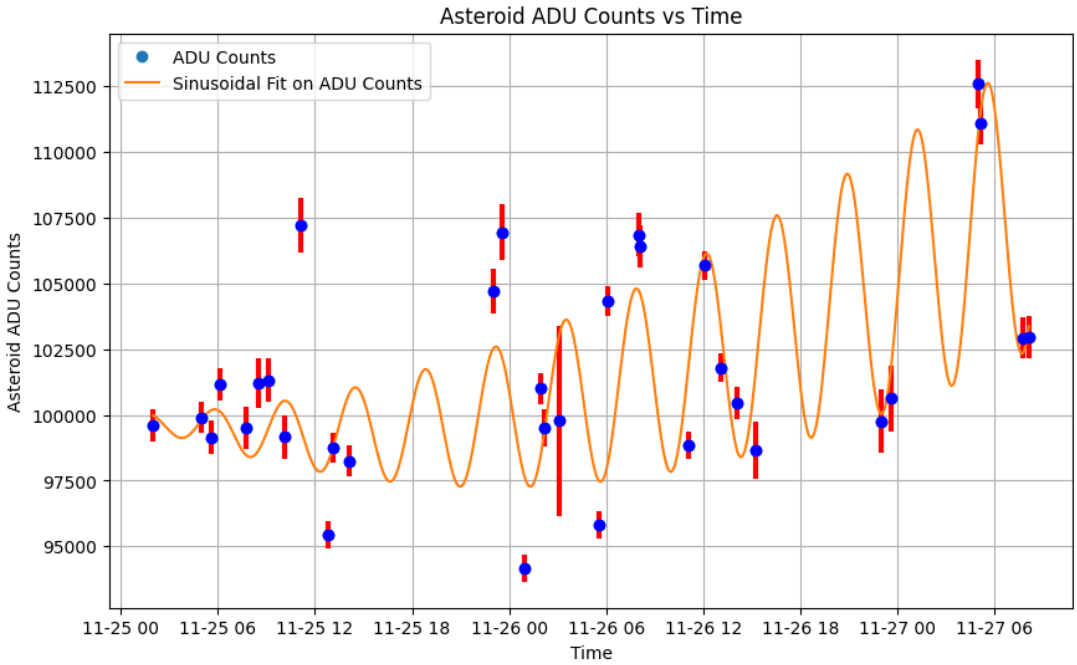

Figure 13: $\chi^2 \approx 416, T_{spot} \approx 4.34h$,$T_{prec} \approx 281h$

A fit of 4.1946 hours was obtained with the **tri-axial ellipsoid with non-homogeneous albedo** model (with linear growth), which is the best fit in terms of the χ^2 statistic and quite close to empiricism.

The fit is: $\chi^2 \approx 169$, $T \approx 4.1946h$, $A \approx 4385$, $B \approx 4015$, $C \approx -1386$, $\phi_1 \approx 0.63rad$, $\phi_2 \approx -1.25rad$

Note that the second and third harmonic amplitude terms are proportional to $\Delta = \frac{c^2 \sin^2 \theta (b^2 - a^2)}{2} < 0$ because these are the terms generated by interaction with the shape model ($a > b > c$).

The second amplitude B's sign has an additional degree of freedom due to it's phase ϕ_2. However, the third amplitude C is clearly negative with constrained peroid. This shows very high significance of real interaction.

5 Summary and Conclusion

This physical analysis successfully recovered a primary rotation period for the target asteroid of around 4.2 hours across multiple optimization runs on different models. These values are in excellent agreement with the empirical literature value of 4.196 h, validating the sensitivity of our periodogram and iterative fitting routines to the dominant signal. However, despite the accuracy of the period and coincidence, the physical validity of the full model—specifically the precession and shape parameters—remains constrained by data quality. The light curve is severely undersampled, leading to degeneracies in the fit.

Although we are confident given our data, that the asteroid is likely a tri-axial ellipsoid with non-homogenous albedo and minimal precession, this model can be invalidated with higher quality data. **We are inconclusive with our results.**

To verify the complex rotation model (Precession + Spot + Shape), future observations must prioritize higher cadence photometry. We recommend a sampling rate of at least 30 data points per 4-hour rotation to satisfy the accuracy for the harmonics and interactions. Furthermore, future analysis should transition from deterministic least-squares fitting (`astropy curve_fit`) to Markov Chain Monte Carlo (MCMC) methods, which allows more robust exploration of the parameter space, easily visualizing and quantifying the covariance between the amplitudes and phases, and providing realistic uncertainty estimates for the asteroid's physical properties.

For the spectral classification, we determine that 16 Psyche is a C,V, or X class asteroid. Without the SDSS Z-band filter, we **cannot make conclusions** due the the requirement of the z-i color index. However, this data agrees with empiricism, which puts the asteroid on the X class.

References

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`https://ssd.jpl.nasa.gov/horizons/app.html#/`

[7] Wikipedia, 16 Psyche Wikipedia,

https://en.wikipedia.org/wiki/16_Psyche

All code can be referenced at: https://github.com/ryanchen2134/lightcurve-analysis
The following link contains all photos used for the lightcurve analysis:
https://observe.lco.global/requestgroups/2387591
The following were used for spectral analysis

- https://observe.lco.global/requests/4034304
- https://observe.lco.global/requests/4034308
- https://observe.lco.global/requests/4034309